

Project II

- **Problem**

- Data: HetIONet (nodes.tsv and edges.tsv)
- Q1: For each drug, compute the number of genes and the number of diseases associated with the drug. Output results with top 5 number of genes in a descending order.
- Q2: Compute the number of diseases associated with 1, 2, 3, ..., n drugs. Output results with the top 5 number of diseases in a descending order.
- Q3: Get the name of drugs that have the top 5 number of genes. Out put the results.

Project II

- **Example**
- Data: HetIONet (nodes.tsv and edges.tsv)

• Q1:

	#genes	#diseases
Drug1	2	1
Drug2	1	2
Drug3	0	1

nodes.tsv

```
Drug1    MagicPill1
Drug2    MagicPill2
Drug3    MagicPill3
```

- Q2: 1 drug -> 2 diseases
2 drugs -> 1 diseases

- Q3: MagicPill1 -> 2
MagicPill2 -> 1
MagicPill3 -> 0

edges.tsv

```
Drug1    G1
Drug1    G2
Drug2    G3
Drug1    Dis1
Drug2    Dis2
Drug2    Dis3
Drug3    Dis4
```

Project II

- **Report**
 - Design of MapReduce algorithm
 - design pattern?
 - Pseudo codes
 - Results for Q1, Q2, and Q3

Project II

- Rubric
 - Distributed algorithm design and implementation
 - Q1: 30 points (15 design + 15 implementation)
 - Q2: 30 points (15 design + 15 implementation)
 - Q3: 20 points (10 design + 10 implementation)
 - Report: (20 points)
 - Must present during your project review!

Project II

- Individual Project
- Due Date
 - 5:00 pm, 11/14